Settling informal claims about modal agents in Chicken

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1 Introduction

Setting

Modal agents play one-shot games by asking what can be proved about each other, written with the provability box $\Box$ of the logic GL [1]. Two informal posts made claims about specific agents that were never checked. "Modal Chicken" [12] introduced CarefulBot and AstuteBot for the game of Chicken, and its comments proposed graded variants: $C_n = \neg\Box(\Box^n\bot \vee p)$ ("swerve unless the opponent provably swerves", with proof strength n) and BoldBots $B_n = \Box(\Box^n\bot \vee \neg p)$, the name a commenter gave to the post's original CarefulBot (B_0); here p means "the opponent swerves against me". "Yet More Modal Combat" [13] proposed, for the prisoner's dilemma, a bot that defects if a weak system proves the opponent cooperates and otherwise acts like FairBot with a strong system; in graded form $H_{a,b} = \neg P_a(p) \wedge P_b(p)$ with $P_a(z) = \Box(\Box^a\bot \vee z)$, $a < b$.

Result

- **Graded CarefulBots and BoldBots, for all grades:** C_n swerves against C_m exactly when $n \leq m$, while B_n swerves against B_m exactly when $n > m$. A CarefulBot against a BoldBot always goes straight while the BoldBot swerves, so the pair always ends at payoffs $(4, 1)$, whatever the grades.
- **The named agents:** a certified table of all 64 pairings of SwerveBot, NerveBot, the original CarefulBot B_0, C_0, AstuteBot as displayed in the post, the FairBots $P_0(p) = \Box p$ and $P_1(p)$, and Hobson's bot $H_{0,1}$ (which cooperates with itself but ends in mutual defection with both FairBots). The original CarefulBot crashes against itself, as a comment pointed out; the displayed AstuteBot swerves against itself, as the post claimed.
- **Hobson's bot, for all grades:** $H_{a,b}$ and $H_{c,d}$ end in mutual cooperation exactly when their weak grades agree ($a = c$), and otherwise in mutual defection. Against any FairBot of the form $P_e(p)$ it ends in mutual defection, for every e.

Why it matters

Informal claims about small agents are easy to make and easy to get wrong; two of the claims in these posts were already disputed in their comment threads. These results settle them for every grade at once. They also show that Hobson's bot, proposed as a bot that cooperates with FairBots, does not cooperate with any graded FairBot of this form.

Proved vs checked

The graded laws (CarefulBots, BoldBots, the mixed pair, Hobson's bot) are proved for all indices in Lean 4, in the height semantics of closed GL fixed points (see [8]; [17]). The 64-cell table is certified by GL fixed-point computation.

Novelty statement

This result extends the informal claims about CarefulBot, AstuteBot and graded CarefulBots and BoldBots in "Modal Chicken" and its comments (Gram_Stone 2015); Hobson's two-system bot and the counterexample in its comments (Hobson 2021). Our prior-art search (2026-10-20) covered:

- Mennen, "Modal combat for games other than the prisoner's dilemma" (LessWrong / Alignment Forum, 2017): framework for modal agents in games with any number of players and actions (player types, ranks, opponent tests naming a target player, an agent to test against and an action); existence and Pareto theorems; no agent families with index laws, no enumeration.
- Gram_Stone, "Modal Chicken" (LessWrong, 2015) and its comments (Alex-Mennen, SilentCal): SwerveBot, NerveBot, CarefulBot, AstuteBot; informal claims about graded CarefulBots and BoldBots.
- Hobson, "Yet More Modal Combat" (LessWrong / Alignment Forum, 2021): a two-system PD bot; generalisation to Chicken and more players left open.
- Critch, Dennis & Russell, "Cooperative and uncooperative institution designs: surprises and problems in open-source game theory" (arXiv 2208.07006): two-player bounded proof-search agents.
- Todorov, ten Napel & Müller, "Parametric Open Source Games" (arXiv 2606.27068): non-modal parametric agents; Chicken and Stag Hunt thresholds.
- Barasz et al. 2014 (arXiv 1401.5577); LaVictoire et al. 2014; Critch, parametric bounded Löb (arXiv 1602.04184); Oesterheld 2022 (arXiv 2211.05057); LessWrong / Alignment Forum wiki "Modal combat"; web searches for modal or provability-based agents in Chicken, Stag Hunt and multiplayer program equilibrium.

If you know of related work we missed, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was partly proved in Lean (claims K5, K11, K12) and partly checked by script (claims K6); the proof and check files are included so anyone can re-run them.

2 Claims

Claim K5

For all $n, m \geq 0$, $C_n = \neg P_n(p)$ swerves against C_m iff $n \leq m$, while $B_n = P_n(\neg p)$ swerves against B_m iff $n > m$, where $P_n(a) = \Box(\Box^n \bot \vee a)$. Exact height laws are Lean-proved.

Status: derived.

Verified.

- `checks/K5-informal_tables.py`: 50 graded pair outcomes for indices 0..4, each with both simultaneous GL fixed-point equations certified. — pass
- `checks/K5-graded_families.py`: Lean core proves exact height laws for every n,m,h and every solution, plus boxed-formula bridge; no sorryAx; false tie law rejected; 20480 finite recurrence cells agree. — pass (Lean-proved, no sorryAx)

Prior art. Partial prior art.

Claim K6

The 64-cell GL table for $S, N, B_0, C_0, A, H, F_0, F_1$ is certified: original CarefulBot B_0 self-crashes; displayed AstuteBot A self-swerves; $H = \neg\Box p \wedge \Box(\Box\bot \vee p)$ self-cooperates but mutually defects with F_0, F_1.

Status: derived.

Verified.

- `checks/K5-informal_tables.py`: All 64 stated outcomes, both GL fixed-point equations per cell, rejection of flipped candidates, and closed-form tail evaluation. — pass
- `checks/K5-informal_tables.py`: 64 outcomes and both simultaneous GL equations per pair, flipped controls rejected; no calculation changed. — pass

Prior art. Partial prior art.

Claim K11

For all $n, m \geq 0$, $C_n = \neg P_n(p)$ goes straight and $B_m = P_m(\neg p)$ swerves against it at every height, where $P_k(a) = \Box(\Box^k \bot \vee a)$. The mixed pair always yields Chicken payoffs $(4, 1)$, independent of grades (Lean).

Status: derived.

Verified.

- `checks/K11-mixed_graded.py`: Lean proves mixed_all for all natural indices/heights and every solution, with no axioms; boxed-formula bridge; 25 GLS cells and 10240 recurrence cells pass; false conclusion rejected. — pass

Prior art. Partial prior art.

Claim K12

For $H_{a,b} = \neg P_a(p) \wedge P_b(p)$, $P_a(z) = \Box(\Box^a \bot \vee z)$, all $a < b, c < d$ give eventual mutual action 1 iff $a = c$, else mutual action 2. Every pairing with $P_e(p)$ eventually gives mutual action 2, for all e (Lean).

Status: derived.

Verified.

- `checks/K12-hobson_grades.py`: 100 generalized Hobson pair cells plus 50 graded FairBot cells; both fixed-point equations GL-proved per cell, flipped candidates rejected. — pass
- `checks/K12-hobson_all_indices.py`: Copied another researcher proof: Lean hob_eq, hob_tail, hob_fair for all indices, boxed bridge, no sorryAx; two false variants rejected; 7098 recurrence cells. — pass (Lean-proved, no sorryAx)

Prior art. Partial prior art.

Provenance. Run `20261019-1400`, claims K5, K6, K11, K12, run-folder commit `24c2a64bf8b8cb8be184b4d449ce11068d8270ea`, published 2026-10-20.

3 Discussion

The results depend on reading each informal definition as a specific modal formula; where a post was ambiguous we state the formula we used, and a different

reading could change the outcome. Hobson's proposal allows other pairs of systems than the graded ones here, so the negative result covers this graded family, not every possible instantiation.

4 How to verify

Each finding has its check scripts in `claims/chicken-informal-claims/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] Christoph Benzmüller. A (simplified) supreme being necessarily exists, says the computer: Computationally explored variants of gödel's ontological argument. In *Proceedings of KR 2020*, 2020.

[3] Christoph Benzmüller. A comment on modal collapse and ultrafilters in gödel's ontological argument. arXiv:2608.07578, 2026.

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[5] Christoph Benzmüller and David Fuenmayor. Computer-supported analysis of positive properties, ultrafilters and modal collapse in variants of gödel's ontological argument. arXiv:1910.08955, 2019.

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[7] Christoph Benzmüller, Leon Weber, and Bruno Woltzenlogel Paleo. Computer-assisted analysis of the anderson–hájek ontological controversy. *Logica Universalis*, 2017.

[8] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[9] Andrew Critch. A parametric, resource-bounded generalization of löb's theorem, and a robust cooperation criterion for open-source game theory. *Journal of Symbolic Logic*, 2019.

[10] Andrew Critch, Michael Dennis, and Stuart Russell. Cooperative and uncooperative institution designs: Surprises and problems in open-source game theory. arXiv:2208.07006, 2022.

[11] Melvin Fitting. *Types, Tableaus, and Gödel's God*. Kluwer, 2002.

[12] Gram_Stone. Modal chicken. LessWrong, 2015.

[13] Donald Hobson. Yet more modal combat. LessWrong / AI Alignment Forum, 2021.

[14] Patrick LaVictoire, Benja Fallenstein, Eliezer Yudkowsky, Mihaly Barasz, Paul Christiano, and Marcello Herreshoff. Program equilibrium in the prisoner's dilemma via löb's theorem. In *AAAI Workshop on Multiagent Interaction without Prior Coordination*, 2014.

[15] Alex Mennen. Modal combat for games other than the prisoner's dilemma. LessWrong / AI Alignment Forum, 2017.

[16] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[17] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.